Precision farming with deep learning by verification process with Tapaal model.

Course Tittle: Autonomous System B

SUBMITTED BY	SUBMITTED TO
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Motivation

01

Monitoring Plants

Monitoring plant leaves to identify plant categories.



03

Design model

Design a Tapaal model for managing a good smart farming

02

Accurate Results

I want to find out a state forward results like name of the categories.

04

Verification

Verification all process like deadlock free or not

Precision farming & Deep Learning

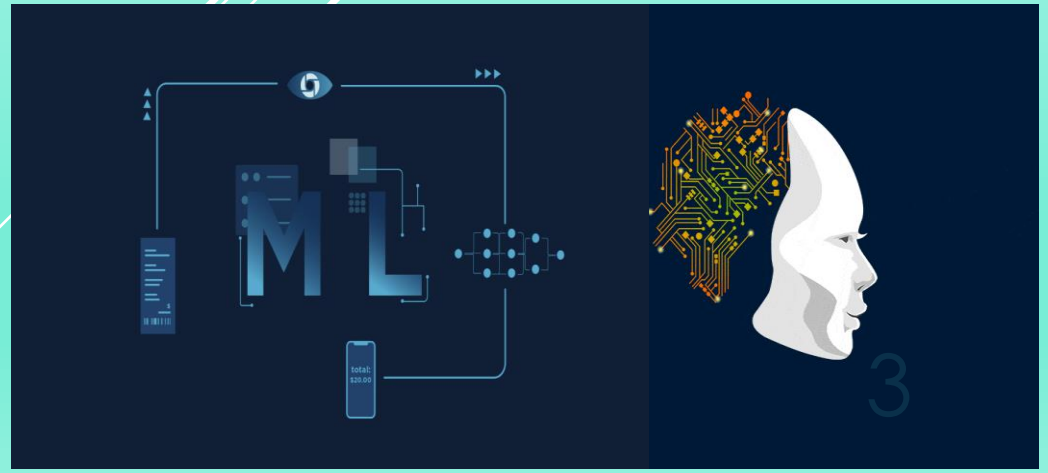
Precision farming :

An agricultural approach that uses technology, data, and tools like drones, sensors, and GPS to monitor and manage targeted farming



Deep Learning:

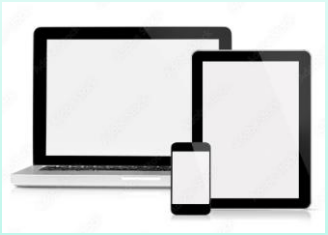
A subset of machine learning that uses neural networks to automatically learn patterns from data



System Design





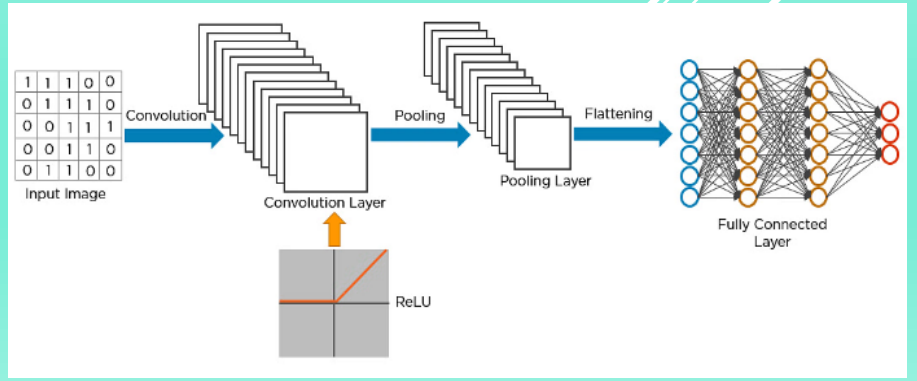


User interface layer





Dataset and Cloud layer



Modelling and Training Layer

CNN Structure

01

Convolution Layer

In the convolution layer according, several feature and filter we get several feature maps for each channel of RGB. Finally, after combining all kernel channel and bias we get final feature map.

02

Residual Blocks layer

This layer use for reduce image size and dimension as well as reduce over feeding since less parameter. There are total two polling method possible including max polling and average polling.

03

Fully connected Layer

This layer is responsible for classifying the images. In the Flatten layer we get one dimensional array and in this layer that is connected with neurons and finally give a output layer with prediction for each class.

CNN Structure

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Convolution

In the convolution layer according, several feature and filter we get several feature maps for each channel of RGB. Finally, after combining all kernel channel and bias we get final feature map. Then for making our feature map non-linear and make training process easy and first we can use Relu (Rectefied Linear Unit).

02

Residual Blocks layer

Residual Blocks enable efficient training of deep networks by using skip connections to pass input directly to the output, helping prevent vanishing gradients and preserving important information..

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CNN Model Summary

```
Total params: 23,532,620
Trainable params: 23,532,620
Non-trainable params: 0
----------------------------------------------------------------
Input size (MB): 0.57
Forward/backward pass size (MB): 286.55
Params size (MB): 89.77
Estimated Total Size (MB): 376.89
----------------------------------------------------------------
```

Dataset

Dataset Source

- Computer Vision in Biosystem

Dataset asset

- No segmented single plant V2
- 12 Plants categories
- 5.5 K Training, Validation & Testing image



Available: https://vision.eng.au.dk/plant-seedlings-dataset/

CNN Implementation

Used tools

- PyTroch
- Sklearn
- Troch vision

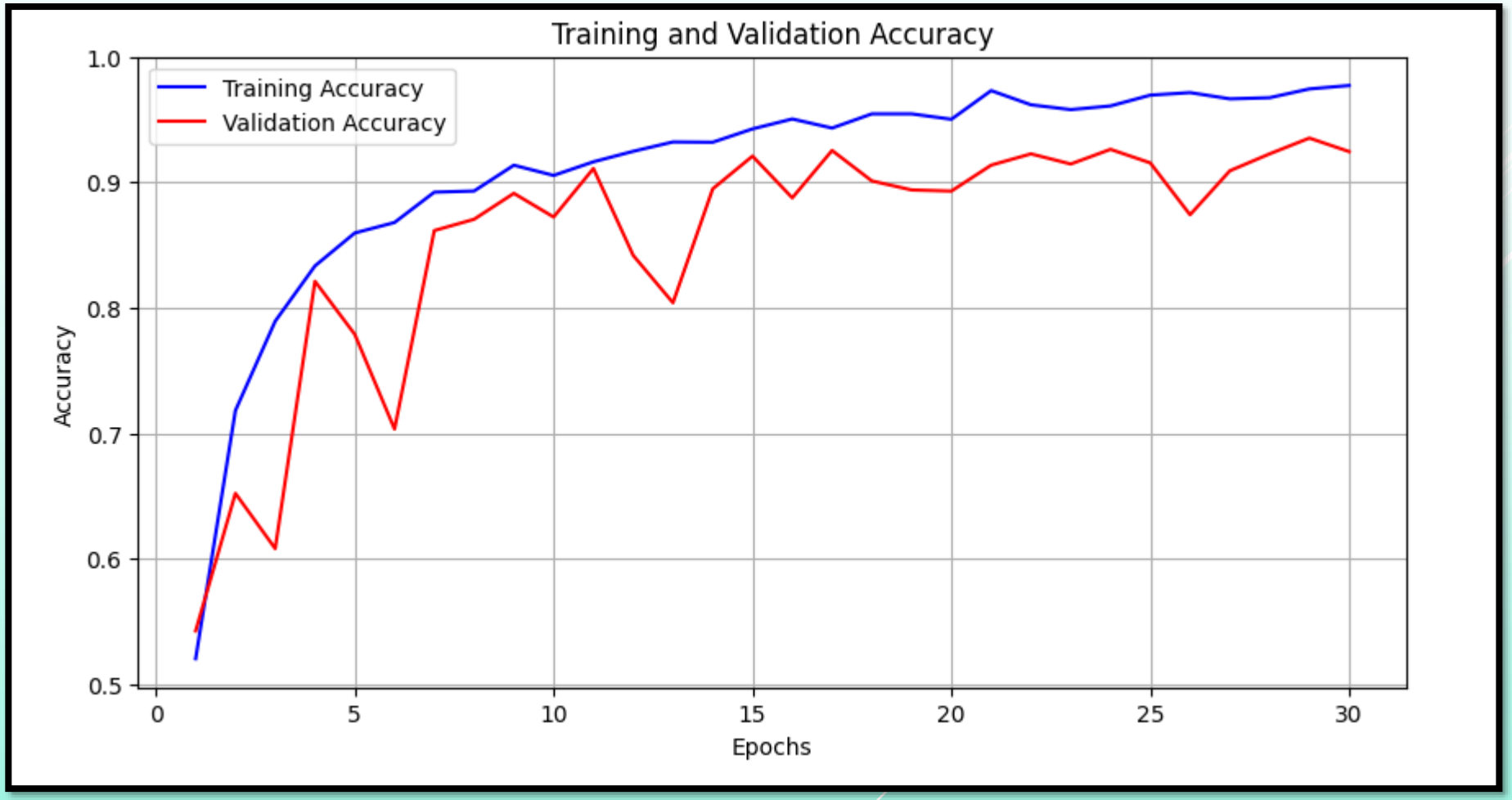
Training Process

- Separate directory for training and validation in 80-20 ration
- Total 30 Epochs
- After getting desire accuracy and losses stopped training

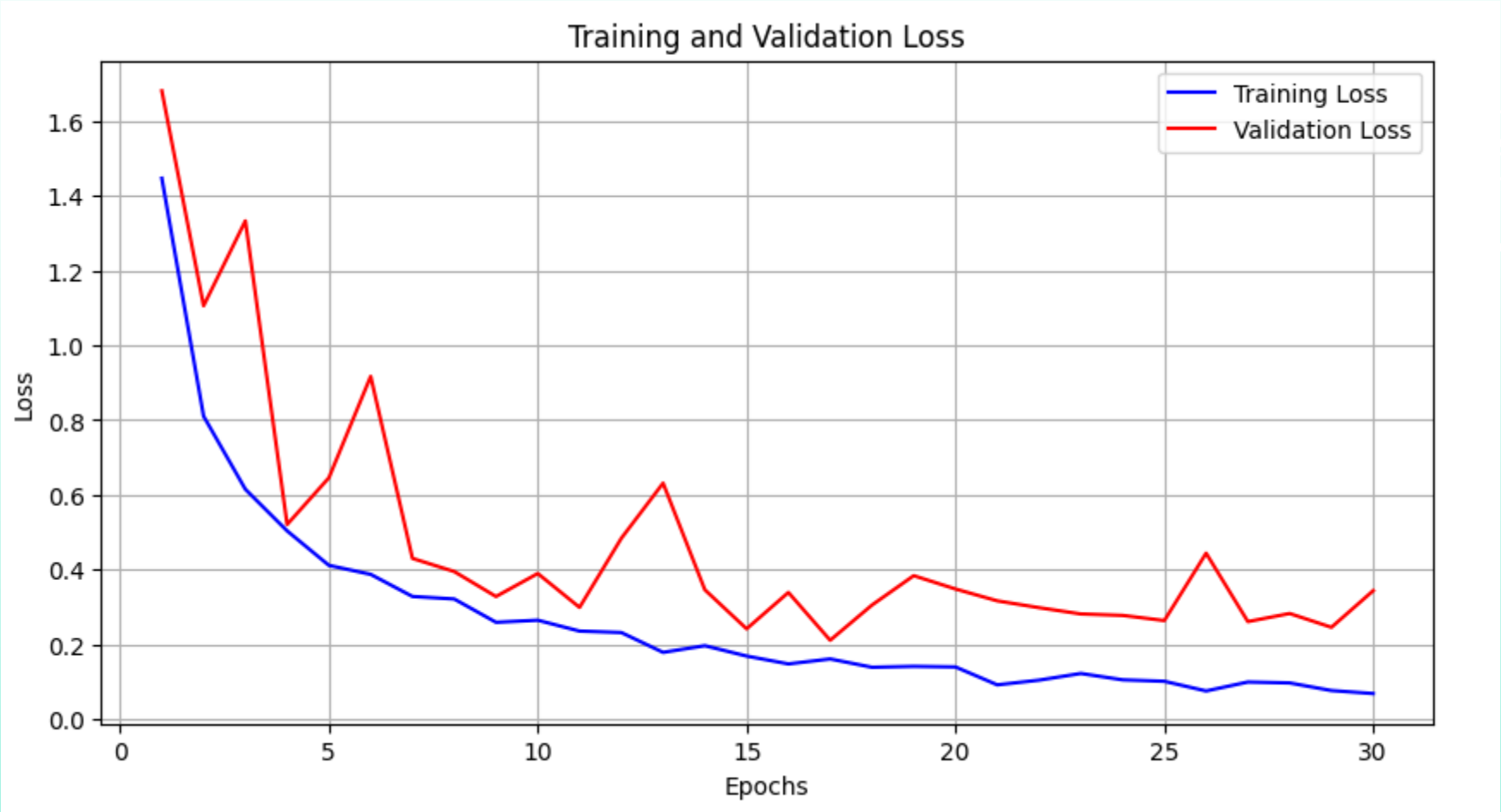
```
Epoch 23/30
----------
train Loss: 0.1223 Acc: 0.9580
val Loss: 0.2814 Acc: 0.9146
Epoch 24/30
----------
train Loss: 0.1053 Acc: 0.9609
val Loss: 0.2776 Acc: 0.9263
Epoch 25/30
----------
train Loss: 0.1014 Acc: 0.9695
val Loss: 0.2638 Acc: 0.9155
Epoch 26/30
----------
train Loss: 0.0755 Acc: 0.9715
val Loss: 0.4439 Acc: 0.8742
Epoch 27/30
----------
train Loss: 0.0994 Acc: 0.9666
val Loss: 0.2610 Acc: 0.9093
Epoch 28/30
----------
train Loss: 0.0970 Acc: 0.9675
val Loss: 0.2826 Acc: 0.9227
Epoch 29/30
----------
train Loss: 0.0766 Acc: 0.9745
val Loss: 0.2457 Acc: 0.9353
Epoch 30/30
----------
train Loss: 0.0686 Acc: 0.9772
val Loss: 0.3437 Acc: 0.9245
Best val Acc: 0.9353
```

Training Epochs

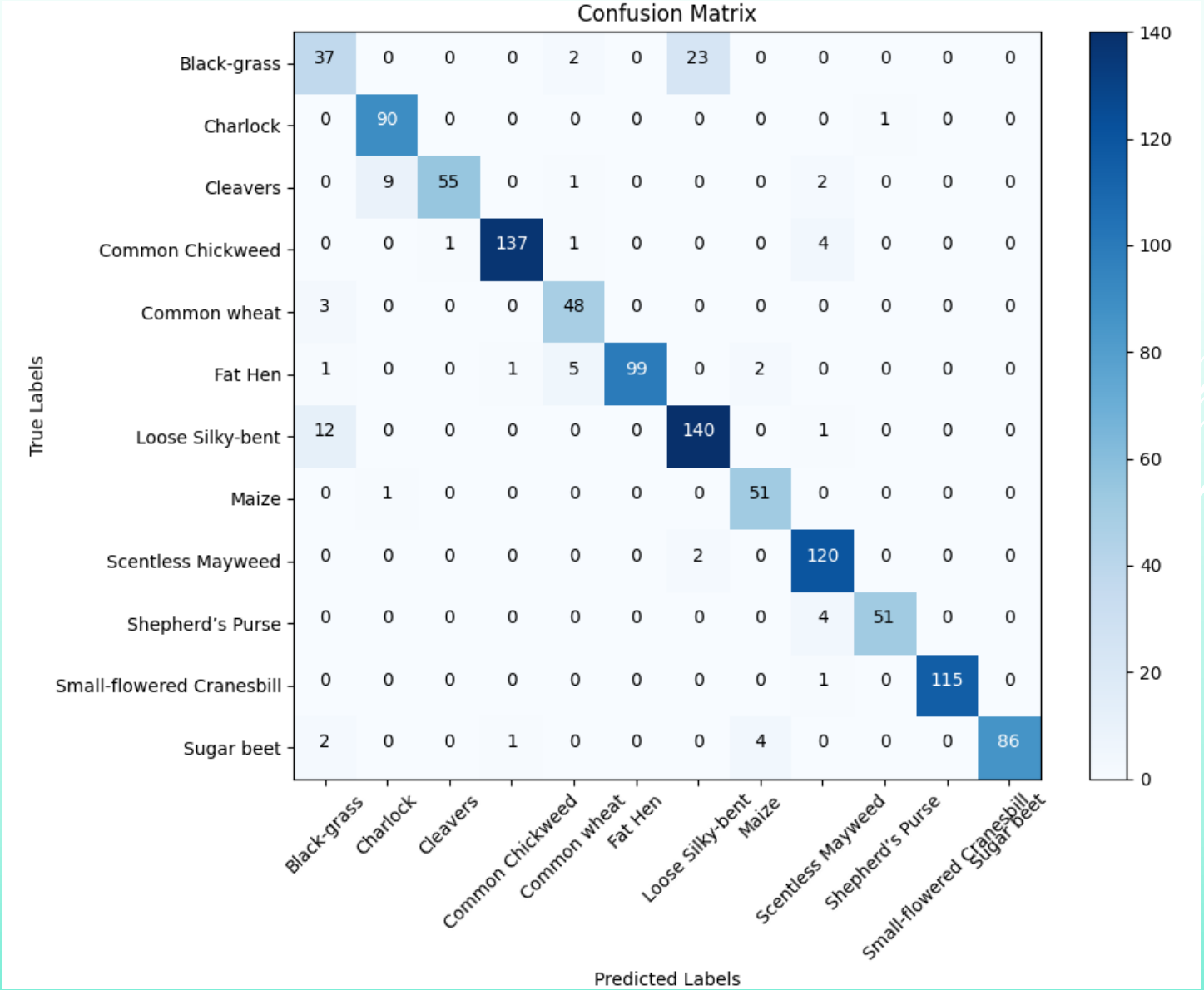
Experimental Evaluation



Experimental Evaluation



Experimental Evaluation



Test model Implementation

Used tools

- Troch vision
- PIL.image
- Trained Model

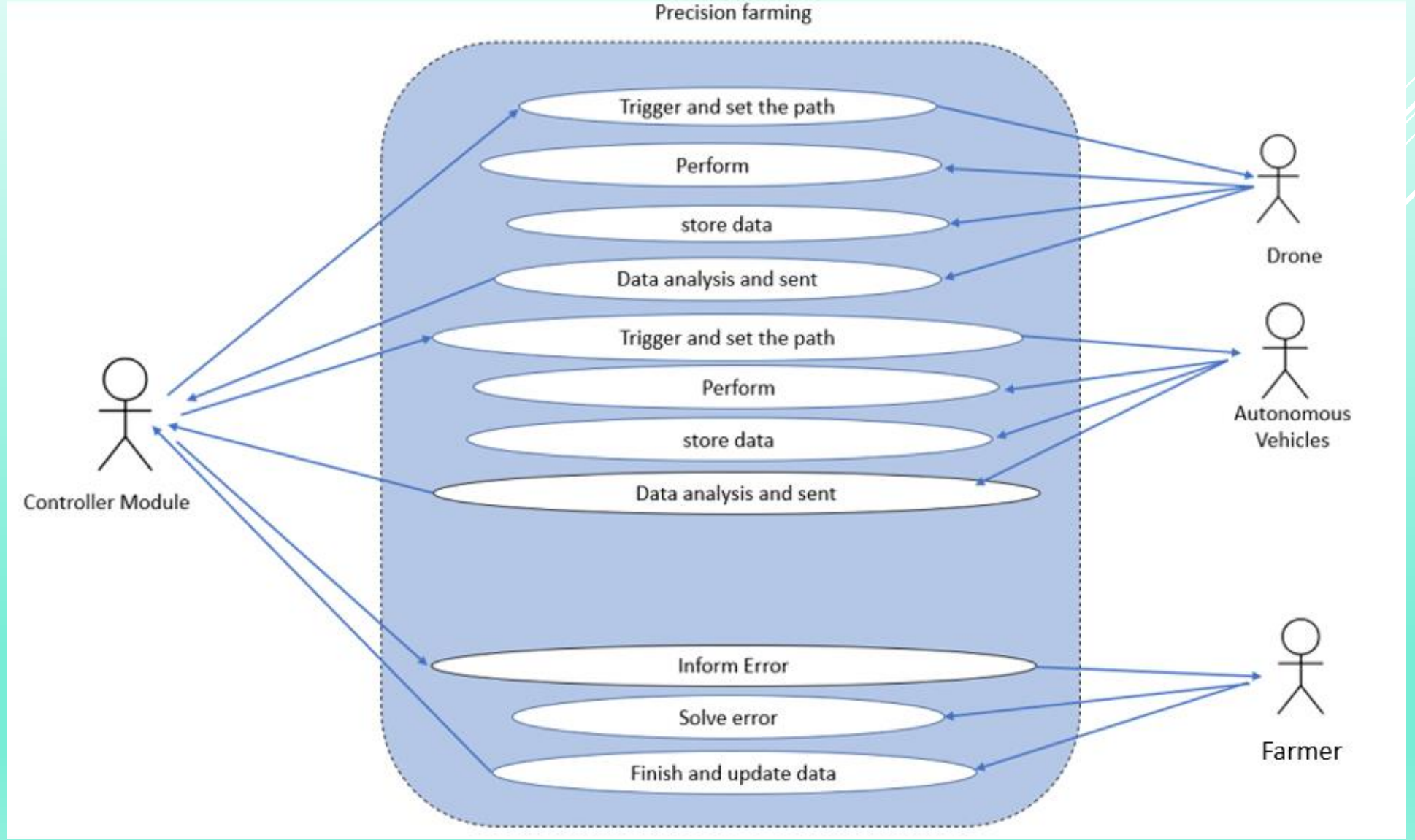
Benefits

- Easy to Use
- Real-Time Results

```
Predicted Class: 11
Confidence: 99.67%
```

```
Dataset sizes: {'train': 4426, 'val': 1113}
Class names: ['Black-grass', 'Charlock', 'Cleavers', 'Common Chickweed', 'Common wheat', 'Fat Hen', 'Loose Silky-bent', 'Maize', 'Scentless Mayweed', 'Sh
epherd's Purse', 'Small-flowered Cranesbill', 'Sugar beet']
```

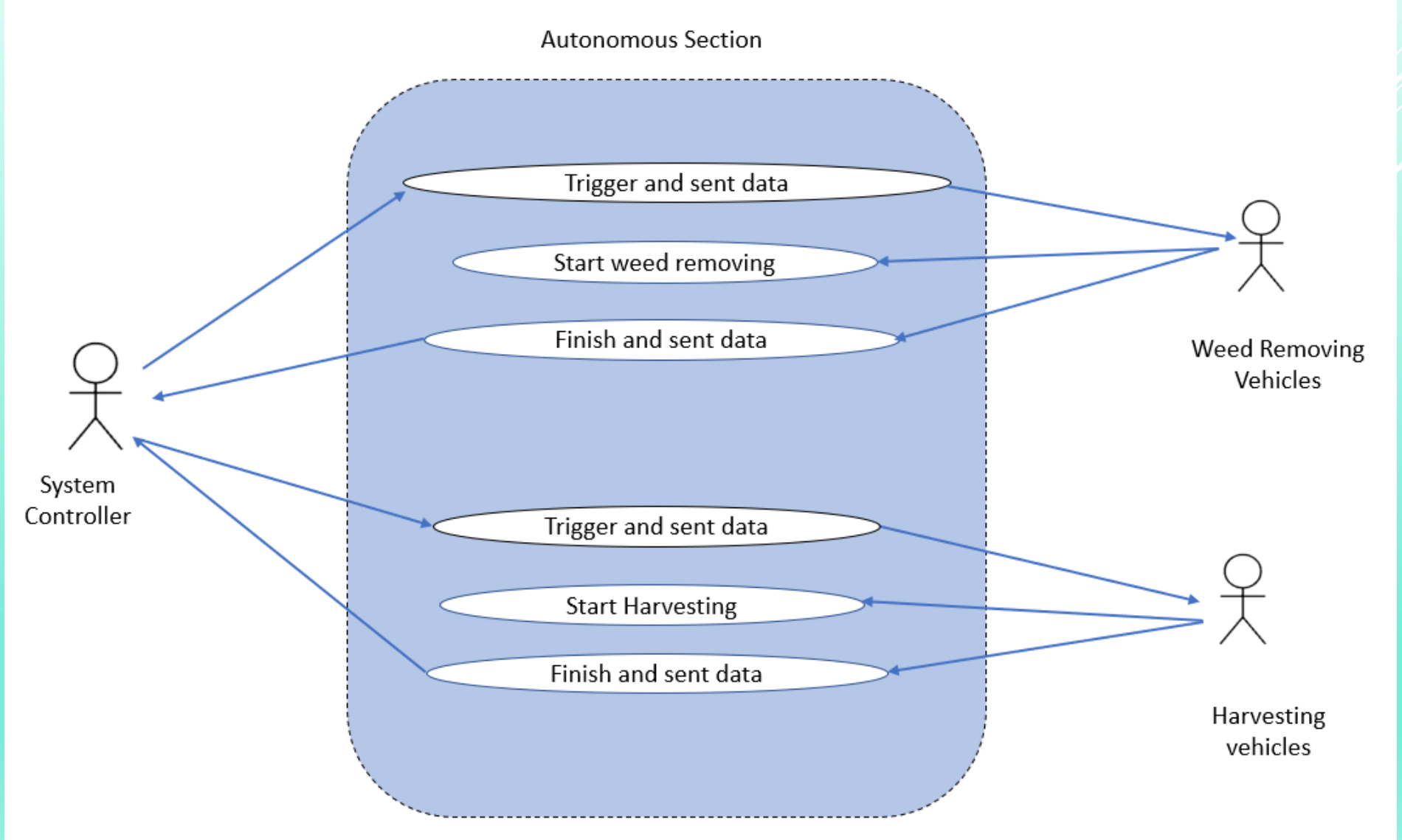
Use Case


Precision farming
Trigger and set the path
Perform
store data
Data analysis and sent
Trigger and set the path
Perform
store data
Data analysis and sent
Inform Error
Solve error
Finish and update data
Drone
Autonomous Vehicles
Farmer
Controller Module

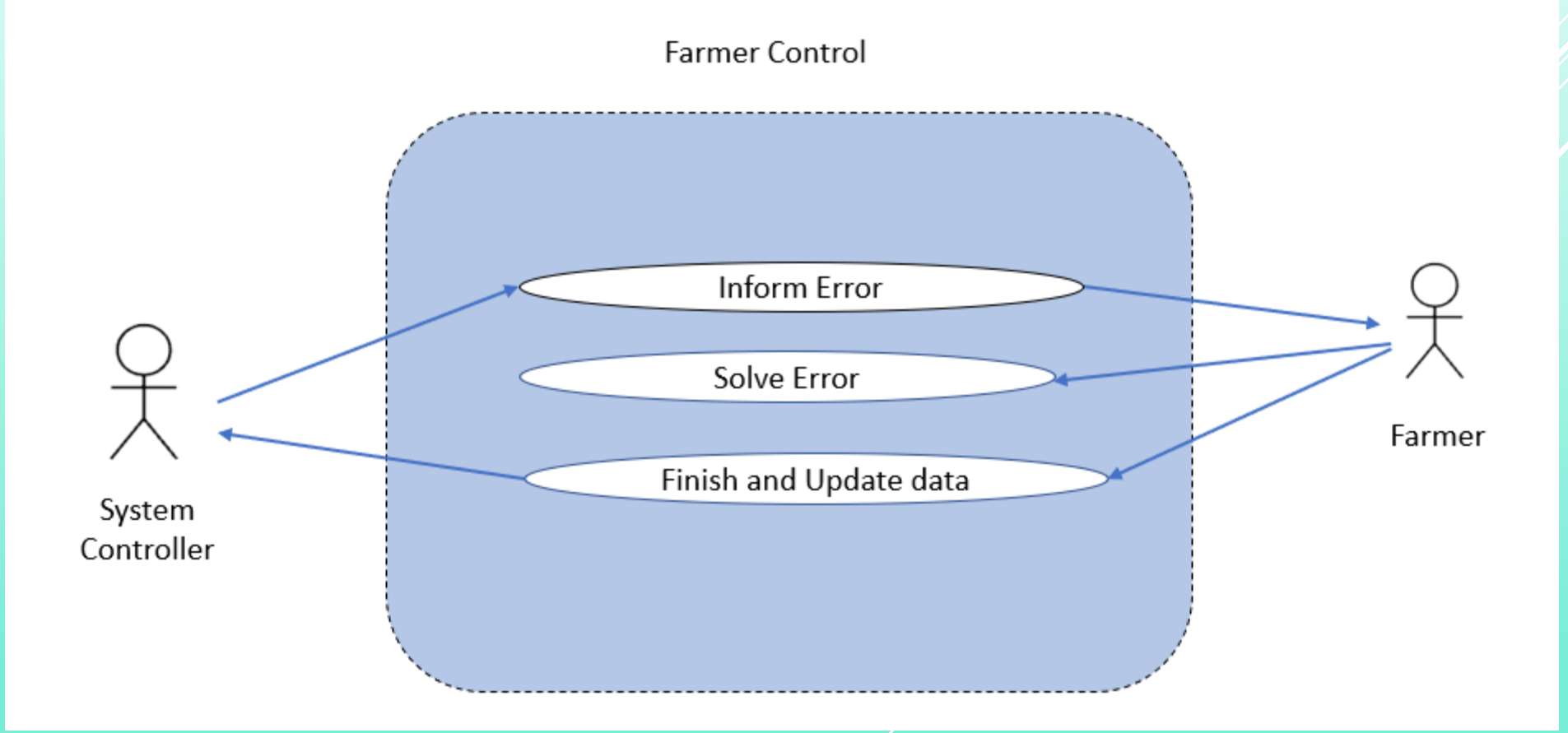
Use Case


Drone Section
Trigger and set the path
Get monitoring data
Monitor and store data
Data analysis and sent
Trigger and sent data
spray pesticide
Finish spray pesticide sent data
Trigger and sent data
Perform Watering
Finish Irrigation and sent data
Trigger and sent data
Collect Weed data
Finish and sent data
Trigger and sent data
Collect Harvesting data
Finish and sent data
System Controller
Survey Drone
Pest control Drone
Watering Drone
Weed Monitoring Drone
Harvesting Monitoring Drone

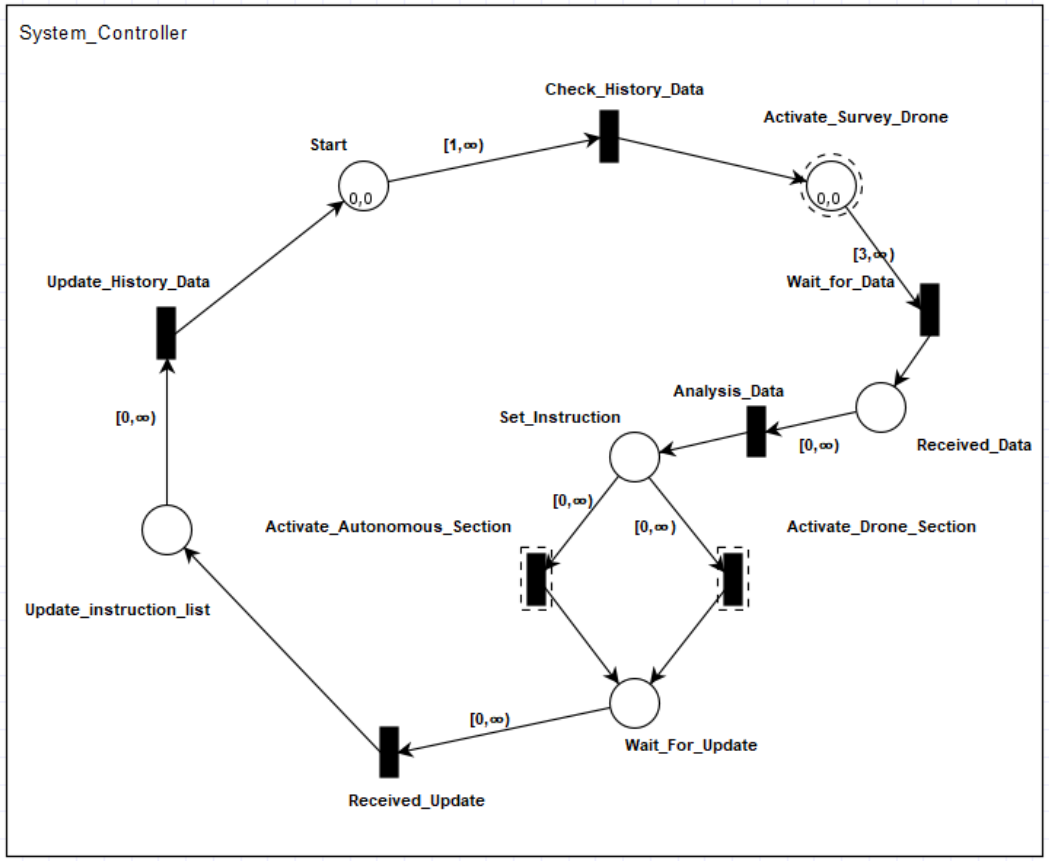
Use Case

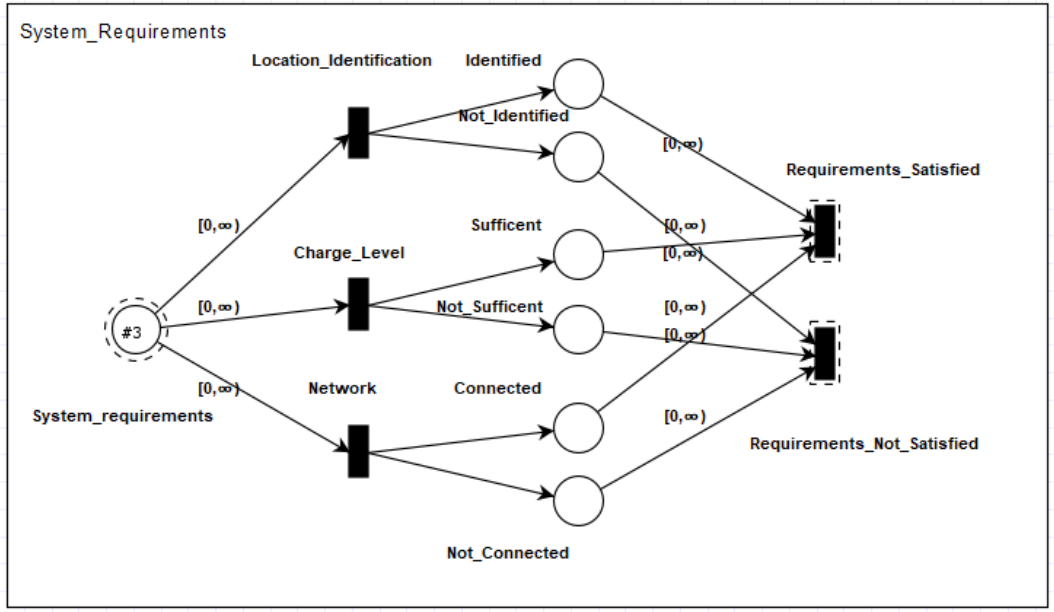

Autonomous Section
Trigger and sent data
Start weed removing
Finish and sent data
Trigger and sent data
Start Harvesting
Finish and sent data
System Controller
Weed Removing Vehicles
Harvesting vehicles

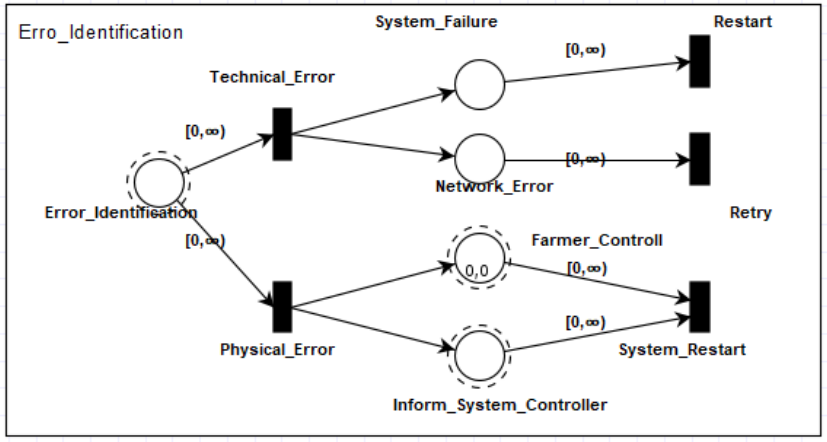
Use Case


Farmer Control
Inform Error
Solve Error
Finish and Update data
System Controller
Farmer

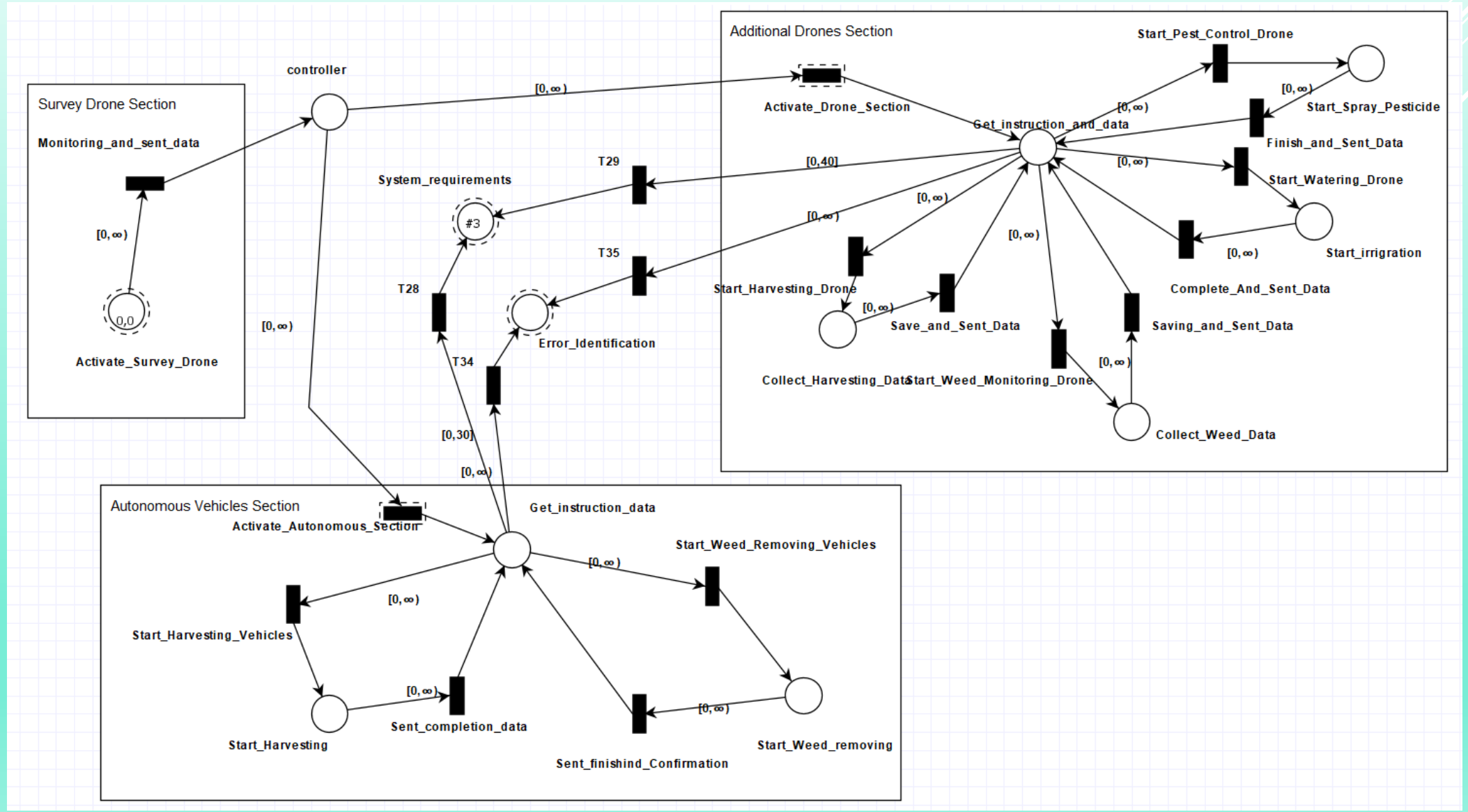
TAPAAL Controller Interface



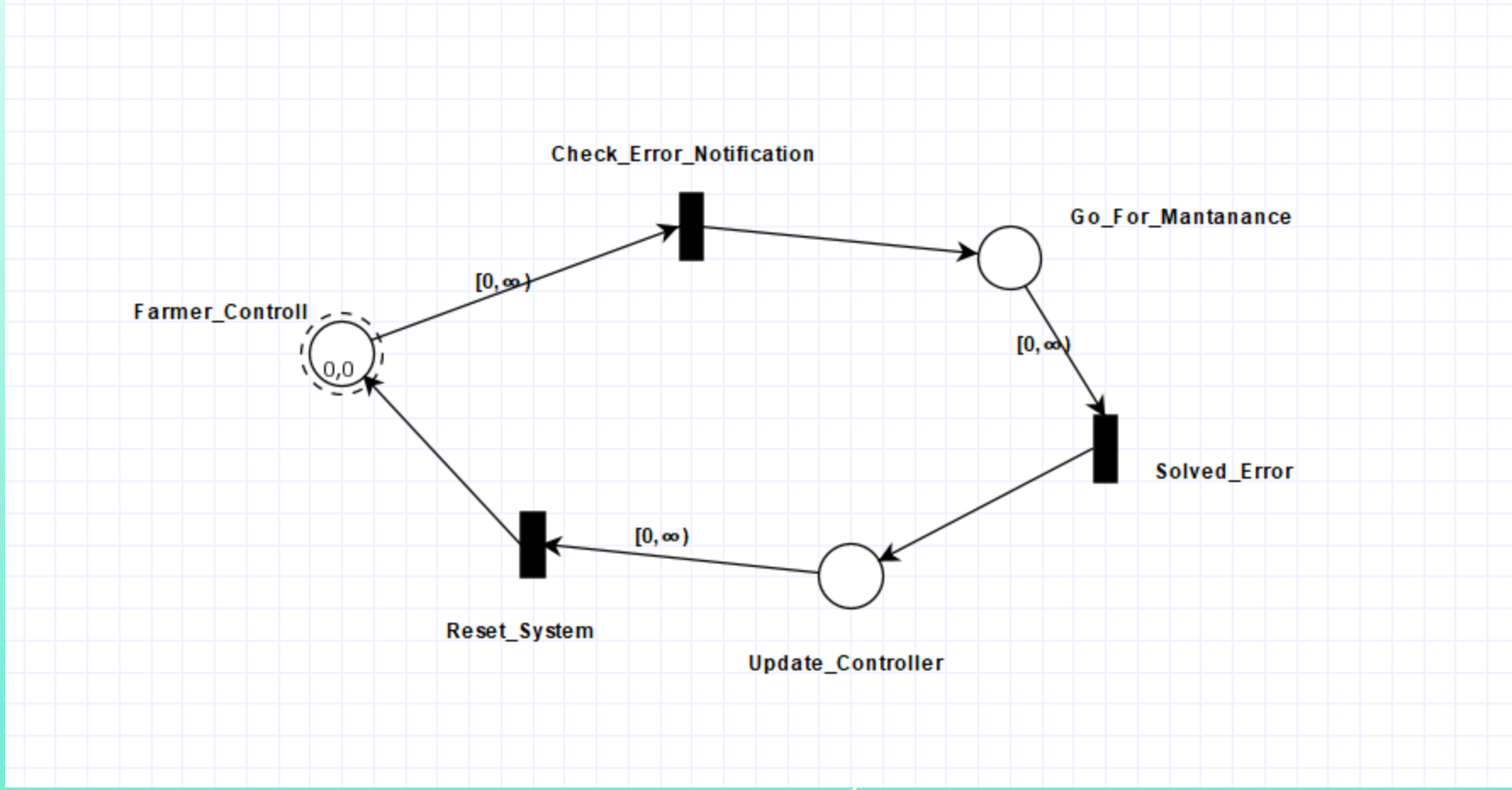




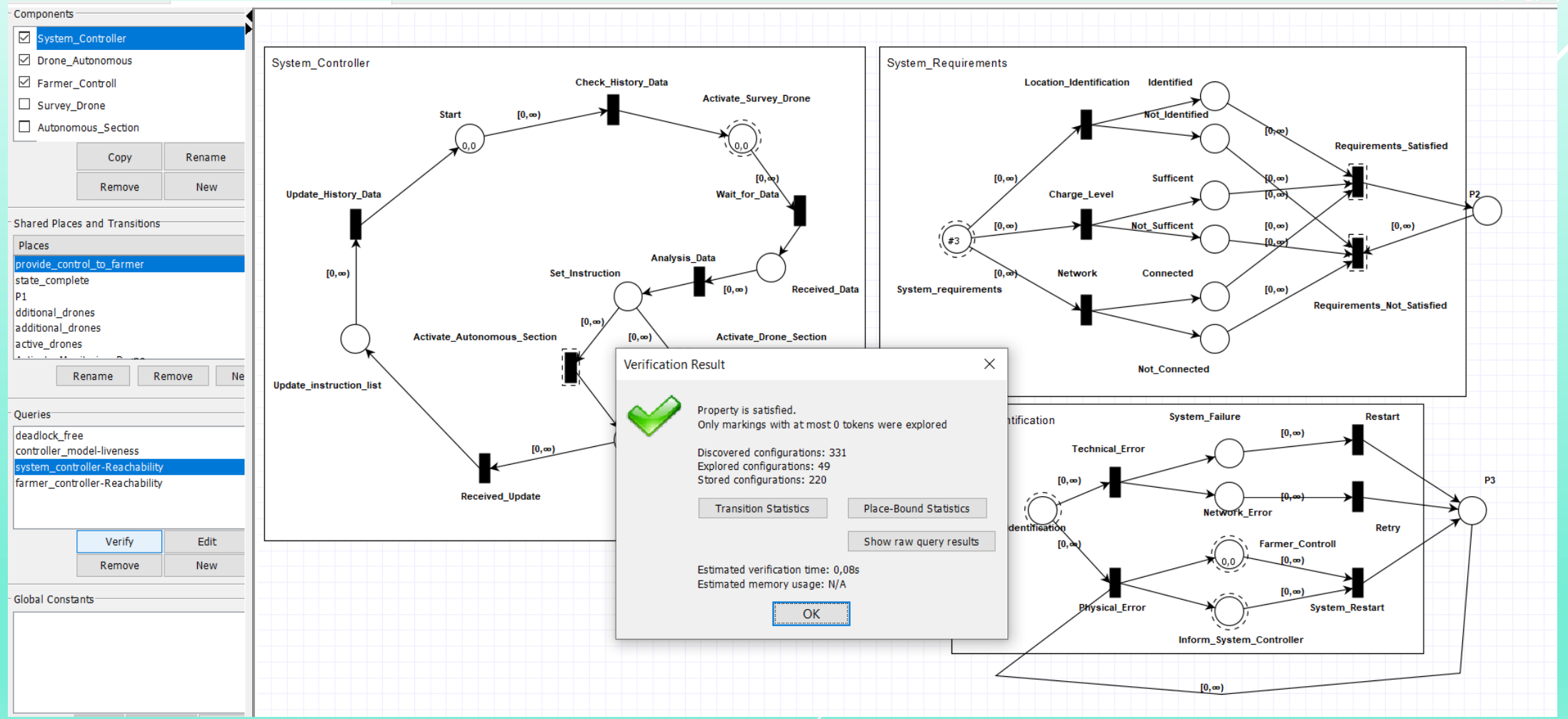
TAPAAL Drone and Autonomous Vehicles Interface



TAPAAL Farmer Control Interface



Verification



Conclusion & Future Work

The System can successfully Identifies plant categories issues with high accuracy

It prove a user friendly interactive testing system

Successfully design a model

I did verify model



Do you have any question?

